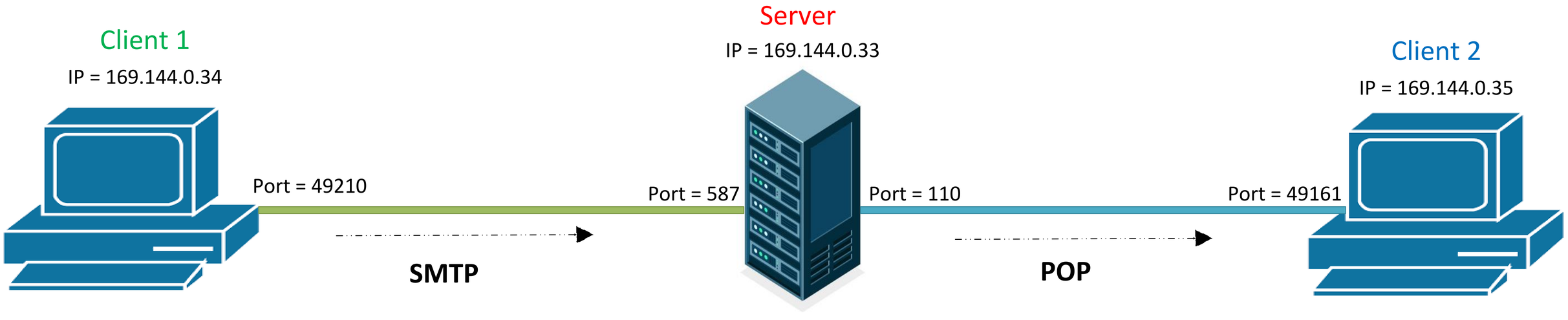

Client 1
IP = 169.144.0.34
Port = 49210
SMTP
Server
IP = 169.144.0.33
Port = 587
Port = 110
POP
Client 2
IP = 169.144.0.35
Port = 49161

# SMTP Table

| # | IPs | IPd | Ps | Pd | Server | Client | Message's on Hexa |
|---|---|---|---|---|---|---|---|
| 1 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 220 WIN-L8MQ2C4F2SC ESMTP | | 32 32 30 20 57 49 4e 2d 4c 38 4d 51 32 43 34 46 32 53 43 20 45 53 4d 54 50 0d 0a |
| 2 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | EHLO [169.144.0.34] | 45 48 4c 4f 20 5b 31 36 39 2e 31 34 34 2e 30 2e 33 34 5d 0d 0a |
| 3 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 250-WIN-L8MQ2C4F2SC 250-SIZE 20480000 250-AUTH LOGIN 250 HELP | | 32 35 30 2d 57 49 4e 2d 4c 38 4d 51 32 43 34 46 32 53 43 0d 0a 32 35 30 2d 53 49 5a 45 20 32 30 34 38 30 30 30 30 0d 0a 32 35 30 2d 41 55 54 48 20 4c 4f 47 49 4e 0d 0a 32 35 30 20 48 45 4c 50 0d 0a |
| 4 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | AUTH LOGIN | 41 55 54 48 20 4c 4f 47 49 4e 0d 0a |
| 5 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 334 VXNlcm5hbWU6 | | 33 33 34 20 56 58 4e 6c 63 6d 35 68 62 57 55 36 0d 0a |
| 6 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | Y2xpZW50MUB0ZXN0bWFpbC5jb20= | 59 32 78 70 5a 57 35 30 4d 55 42 30 5a 58 4e 30 62 57 46 70 62 43 35 6a 62 32 30 3d 0d 0a |
| 7 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 334 UGFzc3dvcmQ6 | | 33 33 34 20 55 47 46 7a 63 33 64 76 63 6d 51 36 0d 0a |
| 8 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | MTIzNDU= | 4d 54 49 7a 4e 44 55 3d 0d 0a |
| 9 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 235 authenticated. | | 32 33 35 20 61 75 74 68 65 6e 74 69 63 61 74 65 64 2e 0d 0a |
| 10 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | MAIL FROM:<client1@testmail.com> SIZE=434 | 4d 41 49 4c 20 46 52 4f 4d 3a 3c 63 6c 69 65 6e 74 31 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 3e 20 53 49 5a 45 3d 34 33 34 0d 0a |
| 11 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 250 OK | | 32 35 30 20 4f 4b 0d 0a |
| 12 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | RCPT TO:<client2@testmail.com> | 52 43 50 54 20 54 4f 3a 3c 63 6c 69 65 6e 74 32 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a |
| 13 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 250 OK | | 32 35 30 20 4f 4b 0d 0a |
| 14 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | DATA | 44 41 54 41 0d 0a |
| 15 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 354 OK, send. | | 33 35 34 20 4f 4b 2c 20 73 65 6e 64 2e 0d 0a |

# SMTP Table

| # | IPs | IPd | Ps | Pd | Server | Client | Message's on Hexa |
|---|---|---|---|---|---|---|---|
| 16 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | **Client**<br>To: client2@testmail.com<br>From: client1 <client1@testmail.com><br>Subject: Test<br>Message-ID: <21bd9d22-54cb-db98-3ab5-5300900f4ba4@testmail.com><br>Date: Thu, 3 Oct 2024 16:42:06 +0300<br>User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64; rv:60.0) Gecko/20100101 Thunderbird/60.9.1<br>MIME-Version: 1.0<br>Content-Type: text/plain; charset=utf-8; format=flowed<br>Content-Transfer-Encoding: 7bit<br>Content-Language: en-US<br><br>Are you there? | | 54 6f 3a 20 63 6c 69 65 6e 74 32 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 0d 0a 46 72 6f 6d 3a 20 63 6c 69 65 6e 74 31 20 3c 63 6c 69 65 6e 74 31 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 53 75 62 6a 65 63 74 3a 20 54 65 73 74 0d 0a 4d 65 73 73 61 67 65 2d 49 44 3a 20 3c 32 31 62 64 39 64 32 32 2d 35 34 63 62 2d 64 62 39 38 2d 33 61 62 35 2d 35 33 30 30 39 30 30 66 34 62 61 34 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 44 61 74 65 3a 20 54 68 75 2c 20 33 20 4f 63 74 20 32 30 32 34 20 31 36 3a 34 32 3a 30 36 20 2b 30 33 30 30 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 36 2e 31 3b 20 57 69 6e 36 34 3b 20 78 36 34 3b 20 72 76 3a 36 30 2e 30 29 20 47 65 63 6b 6f 2f 32 30 31 30 30 31 30 31 0d 0a 20 54 68 75 6e 64 65 72 62 69 72 64 2f 36 30 2e 39 2e 31 0d 0a 4d 49 4d 45 2d 56 65 72 73 69 6f 6e 3a 20 31 2e 30 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f 70 6c 61 69 6e 3b 20 63 68 61 72 73 65 74 3d 75 74 66 2d 38 3b 20 66 6f 72 6d 61 74 3d 66 6c 6f 77 65 64 0d 0a 43 6f 6e 74 65 6e 74 2d 54 72 61 6e 73 66 65 72 2d 45 6e 63 6f 64 69 6e 67 3a 20 37 62 69 74 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d 55 53 0d 0a 0d 0a 41 72 65 20 79 6f 75 20 74 68 65 72 65 3f 0d 0a 0d 0a |
| 17 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | . | 2e 0d 0a |
| 18 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 250 Queued (0.000 seconds) | | 32 35 30 20 51 75 65 75 65 64 20 28 30 2e 30 30 30 20 73 65 63 6f 6e 64 73 29 0d 0a |
| 19 | 169.144.0.34 | 169.144.0.33 | 49210 | 587 | | QUIT | 51 55 49 54 0d 0a |
| 20 | 169.144.0.33 | 169.144.0.34 | 587 | 49210 | 221 goodbye | | 32 32 31 20 67 6f 6f 64 62 79 65 0d 0a |

# POP Table

| # | IPs | IPd | Ps | Pd | Server | Client | Message's on Hexa |
|---|---|---|---|---|---|---|---|
| 1 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK POP3 | | 2b 4f 4b 20 50 4f 50 33 0d 0a |
| 2 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | CAPA | 43 41 50 41 0d 0a |
| 3 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK CAPA list follows<br>USER<br>UIDL<br>TOP<br>. | | 2b 4f 4b 20 43 41 50 41 20 6c 69 73 74 20 66 6f 6c 6c 6f 77 73 0d 0a 55 53 45 52 0d 0a 55 49 44 4c 0d 0a 54 4f 50 0d 0a 2e 0d 0a |
| 4 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | USER client2@testmail.com | 55 53 45 52 20 63 6c 69 65 6e 74 32 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 0d 0a |
| 5 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK Send your password | | 2b 4f 4b 20 53 65 6e 64 20 79 6f 75 72 20 70 61 73 73 77 6f 72 64 0d 0a |
| 6 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | PASS 12345 | 50 41 53 53 20 31 32 33 34 35 0d 0a |
| 7 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK Mailbox locked and ready | | 2b 4f 4b 20 4d 61 69 6c 62 6f 78 20 6c 6f 63 6b 65 64 20 61 6e 64 20 72 65 61 64 79 0d 0a |
| 8 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | STAT | 53 54 41 54 0d 0a |
| 9 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK 2 1148 | | 2b 4f 4b 20 32 20 31 31 34 38 0d 0a |
| 10 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | LIST | 4c 49 53 54 0d 0a |
| 11 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK 2 messages (1148 octets) | | 2b 4f 4b 20 32 20 6d 65 73 73 61 67 65 73 20 28 31 31 34 38 20 6f 63 74 65 74 73 29 0d 0a |
| 12 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | 1 582<br>2 566<br>. | | 31 20 35 38 32 0d 0a 32 20 35 36 36 0d 0a 2e 0d 0a |
| 13 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | UIDL | 55 49 44 4c 0d 0a |
| 14 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK 2 messages (1148 octets)<br>1 1<br>2 2<br>. | | 2b 4f 4b 20 32 20 6d 65 73 73 61 67 65 73 20 28 31 31 34 38 20 6f 63 74 65 74 73 29 0d 0a 31 20 31 0d 0a 32 20 32 0d 0a 2e 0d 0a |
| 15 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | RETR 2 | 52 45 54 52 20 32 0d 0a |

# POP Table

| # | IPs | IPd | Ps | Pd | Server | Client | Message's on Hexa |
|---|---|---|---|---|---|---|---|
| 16 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | **Server**<br><br>+OK 566 octets<br>Return-Path: client1@testmail.com<br>Received: from [169.144.0.34] (WIN-UKBDG8C76RL [169.144.0.34])<br>by WIN-L8MQ2C4F2SC with ESMTPA<br>; Thu, 3 Oct 2024 16:42:06 +0300<br>To: client2@testmail.com<br>From: client1 <client1@testmail.com><br>Subject: Test<br>Message-ID: <21bd9d22-54cb-db98-3ab5-5300900f4ba4@testmail.com><br>Date: Thu, 3 Oct 2024 16:42:06 +0300<br>User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64; rv:60.0) Gecko/20100101 Thunderbird/60.9.1<br>MIME-Version: 1.0<br>Content-Type: text/plain; charset=utf-8; format=flowed<br>Content-Transfer-Encoding: 7bit<br>Content-Language: en-US<br><br>Are you there? | | 2b 4f 4b 20 35 36 36 20 6f 63 74 65 74 73 0d 0a<br>52 65 74 75 72 6e 2d 50 61 74 68 3a 20 63 6c 69 65 6e 74 31 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 0d 0a 52 65 63 65 69 76 65 64 3a 20 66 72 6f 6d 20 5b 31 36 39 2e 31 34 34 2e 30 2e 33 34 5d 20 28 57 49 4e 2d 55 4b 42 44 47 38 43 37 36 52 4c 20 5b 31 36 39 2e 31 34 34 2e 30 2e 33 34 5d 29 0d 0a 09 62 79 20 57 49 4e 2d 4c 38 4d 51 32 43 34 46 32 53 43 20 77 69 74 68 20 45 53 4d 54 50 41 0d 0a 09 3b 20 54 68 75 2c 20 33 20 4f 63 74 20 32 30 32 34 20 31 36 3a 34 32 3a 30 36 20 2b 30 33 30 30 0d 0a 54 6f 3a 20 63 6c 69 65 6e 74 32 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 0d 0a 46 72 6f 6d 3a 20 63 6c 69 65 6e 74 31 20 3c 63 6c 69 65 6e 74 31 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 53 75 62 6a 65 63 74 3a 20 54 65 73 74 0d 0a 4d 65 73 73 61 67 65 2d 49 44 3a 20 3c 32 31 62 64 39 64 32 32 2d 35 34 63 62 2d 64 62 39 38 2d 33 61 62 35 2d 35 33 30 30 39 30 30 66 34 62 61 34 40 74 65 73 74 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 44 61 74 65 3a 20 54 68 75 2c 20 33 20 4f 63 74 20 32 30 32 34 20 31 36 3a 34 32 3a 30 36 20 2b 30 33 30 30 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 36 2e 31 3b 20 57 69 6e 36 34 3b 20 78 36 34 3b 20 72 76 3a 36 30 2e 30 29 20 47 65 63 6b 6f 2f 32 30 31 30 30 31 30 31 0d 0a 20 54 68 75 6e 64 65 72 62 69 72 64 2f 36 30 2e 39 2e 31 0d 0a 4d 49 4d 45 2d 56 65 72 73 69 6f 6e 3a 20 31 2e 30 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f 70 6c 61 69 6e 3b 20 63 68 61 72 73 65 74 3d 75 74 66 2d 38 3b 20 66 6f 72 6d 61 74 3d 66 6c 6f 77 65 64 0d 0a 43 6f 6e 74 65 6e 74 2d 54 72 61 6e 73 66 65 72 2d 45 6e 63 6f 64 69 6e 67 3a 20 37 62 69 74 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d 55 53 0d 0a 0d 0a 41 72 65 20 79 6f 75 20 74 68 65 72 65 3f 0d 0a 0d 0a |
| 17 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | . | | 2e 0d 0a |
| 18 | 169.144.0.35 | 169.144.0.33 | 49161 | 110 | | QUIT | 51 55 49 54 0d 0a |
| 19 | 169.144.0.33 | 169.144.0.35 | 110 | 49161 | +OK POP3 server saying goodbye... | | 2b 4f 4b 20 50 4f 50 33 20 73 65 72 76 65 72 20 73 61 79 69 6e 67 20 67 6f 6f 64 62 79 65 2e 2e 2e 0d 0a |

Client 1

IP = 169.144.0.34

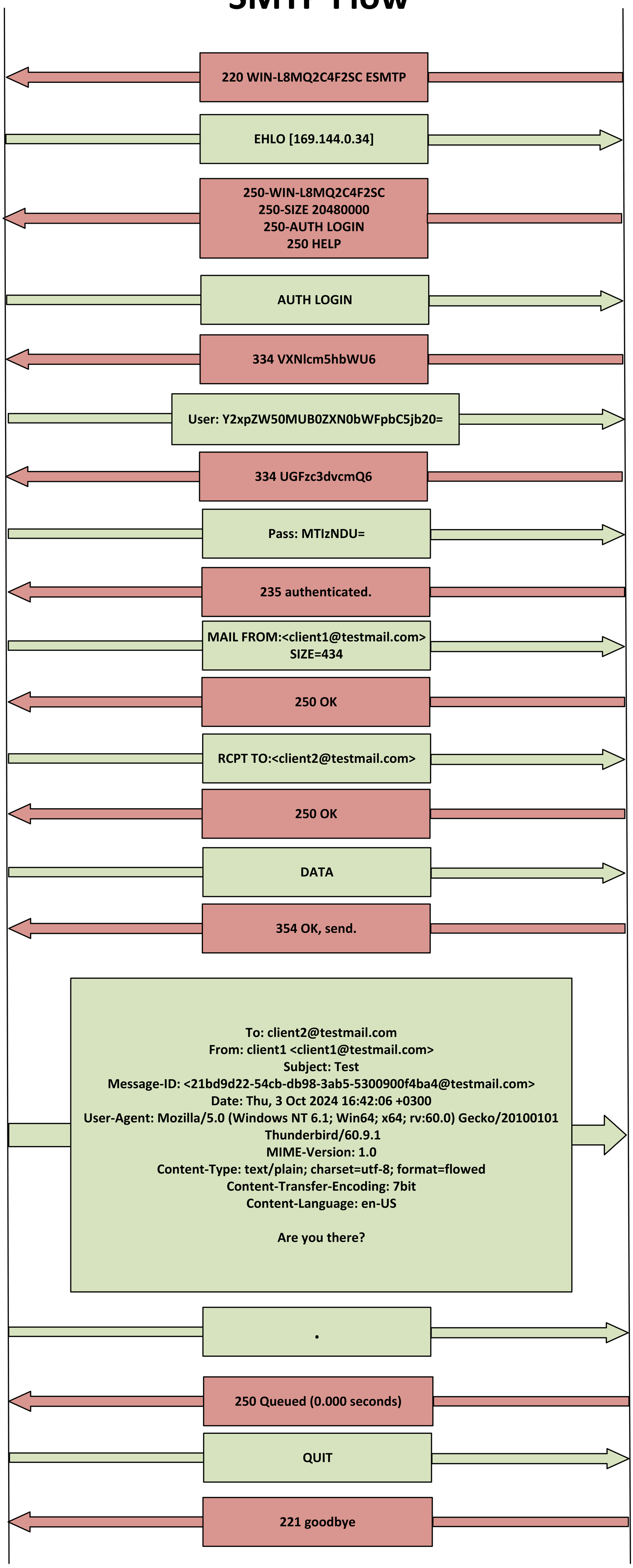
Port = 49210

Server

IP = 169.144.0.33

Port = 587

# SMTP Flow



Client 2

IP = 169.144.0.35

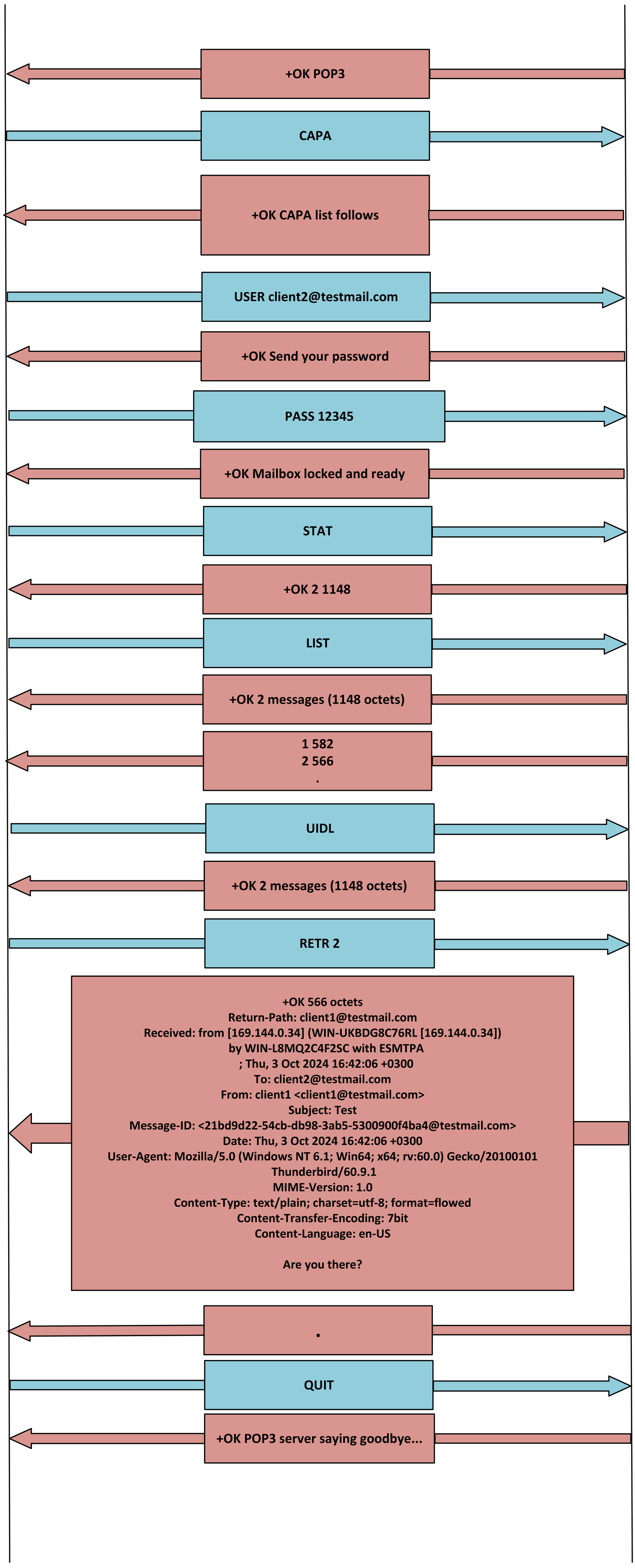
Port = 49161

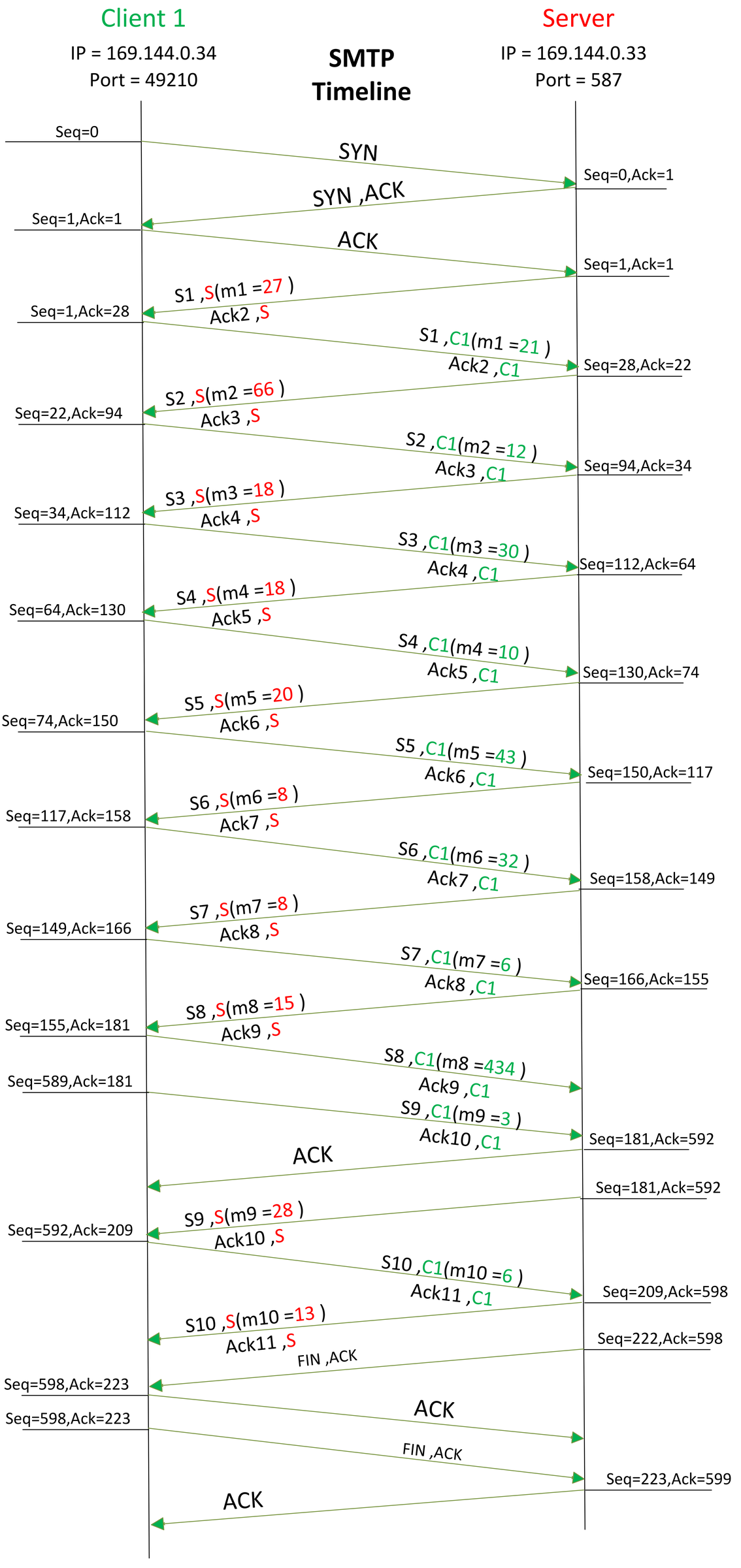
# POP Flow

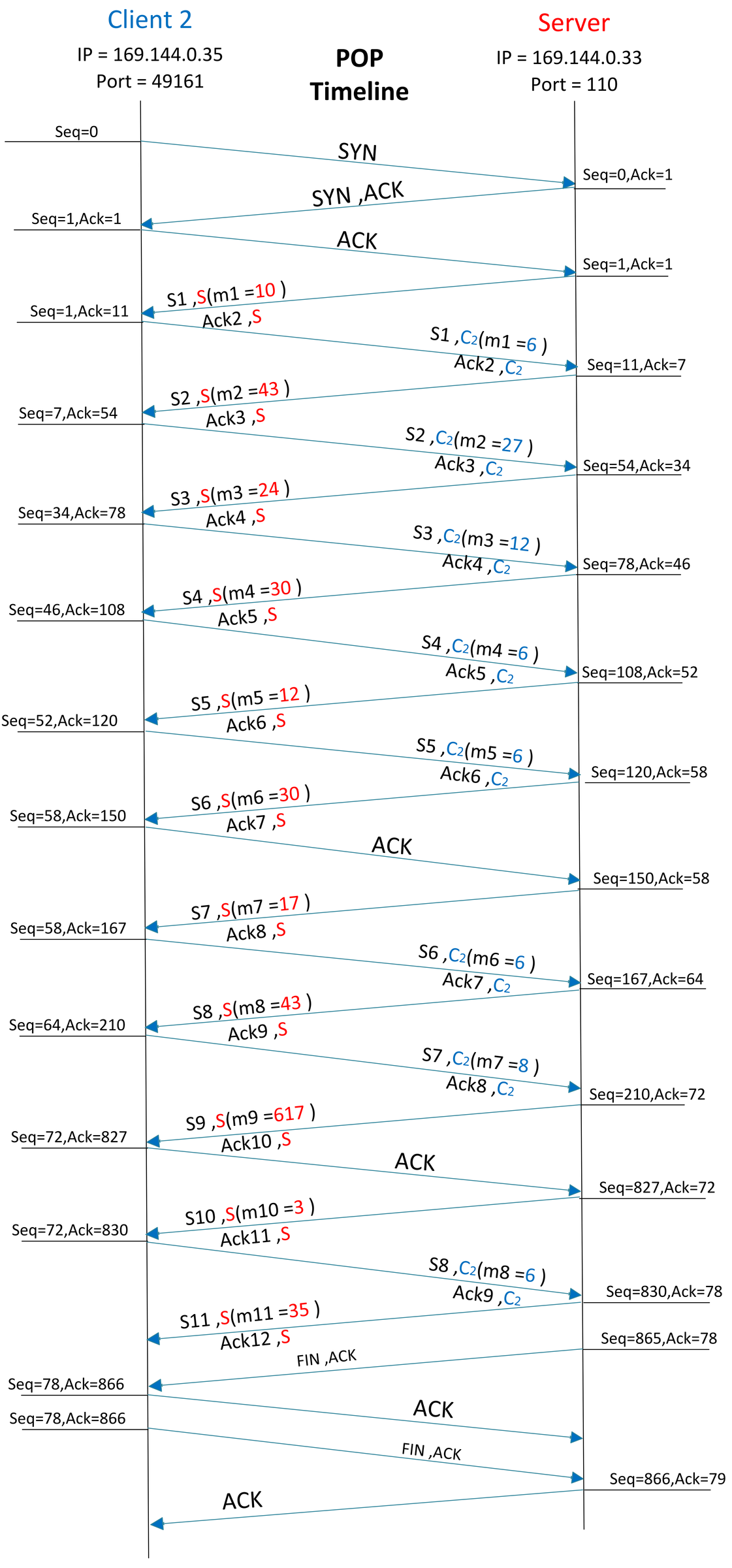
Server

IP = 169.144.0.33

Port = 110




Client 1
IP = 169.144.0.34
Port = 49210
SMTP Timeline
Server
IP = 169.144.0.33
Port = 587
Seq=0
SYN
Seq=0,Ack=1
SYN ,ACK
Seq=1,Ack=1
ACK
Seq=1,Ack=1
S1 ,S(m1 =27 )
Ack2 ,S
Seq=1,Ack=28
S1 ,C1(m1 =21 )
Ack2 ,C1
Seq=28,Ack=22
S2 ,S(m2 =66 )
Ack3 ,S
Seq=22,Ack=94
S2 ,C1(m2 =12 )
Ack3 ,C1
Seq=94,Ack=34
S3 ,S(m3 =18 )
Ack4 ,S
Seq=34,Ack=112
S3 ,C1(m3 =30 )
Ack4 ,C1
Seq=112,Ack=64
S4 ,S(m4 =18 )
Ack5 ,S
Seq=64,Ack=130
S4 ,C1(m4 =10 )
Ack5 ,C1
Seq=130,Ack=74
S5 ,S(m5 =20 )
Ack6 ,S
Seq=74,Ack=150
S5 ,C1(m5 =43 )
Ack6 ,C1
Seq=150,Ack=117
S6 ,S(m6 =8 )
Ack7 ,S
Seq=117,Ack=158
S6 ,C1(m6 =32 )
Ack7 ,C1
Seq=158,Ack=149
S7 ,S(m7 =8 )
Ack8 ,S
Seq=149,Ack=166
S7 ,C1(m7 =6 )
Ack8 ,C1
Seq=166,Ack=155
S8 ,S(m8 =15 )
Ack9 ,S
Seq=155,Ack=181
S8 ,C1(m8 =434 )
Ack9 ,C1
Seq=589,Ack=181
S9 ,C1(m9 =3 )
Ack10 ,C1
Seq=181,Ack=592
ACK
Seq=181,Ack=592
S9 ,S(m9 =28 )
Ack10 ,S
Seq=592,Ack=209
S10 ,C1(m10 =6 )
Ack11 ,C1
Seq=209,Ack=598
S10 ,S(m10 =13 )
Ack11 ,S
Seq=222,Ack=598
FIN ,ACK
Seq=598,Ack=223
ACK
Seq=598,Ack=223
FIN ,ACK
Seq=223,Ack=599
ACK


Client 2
IP = 169.144.0.35
Port = 49161
POP
Timeline
Server
IP = 169.144.0.33
Port = 110
Seq=0
SYN
Seq=0,Ack=1
SYN ,ACK
Seq=1,Ack=1
ACK
Seq=1,Ack=1
S1 ,S(m1 =10 )
Ack2 ,S
Seq=1,Ack=11
S1 ,C2(m1 =6 )
Ack2 ,C2
Seq=11,Ack=7
S2 ,S(m2 =43 )
Ack3 ,S
Seq=7,Ack=54
S2 ,C2(m2 =27 )
Ack3 ,C2
Seq=54,Ack=34
S3 ,S(m3 =24 )
Ack4 ,S
Seq=34,Ack=78
S3 ,C2(m3 =12 )
Ack4 ,C2
Seq=78,Ack=46
S4 ,S(m4 =30 )
Ack5 ,S
Seq=46,Ack=108
S4 ,C2(m4 =6 )
Ack5 ,C2
Seq=108,Ack=52
S5 ,S(m5 =12 )
Ack6 ,S
Seq=52,Ack=120
S5 ,C2(m5 =6 )
Ack6 ,C2
Seq=120,Ack=58
S6 ,S(m6 =30 )
Ack7 ,S
Seq=58,Ack=150
ACK
Seq=150,Ack=58
S7 ,S(m7 =17 )
Ack8 ,S
Seq=58,Ack=167
S6 ,C2(m6 =6 )
Ack7 ,C2
Seq=167,Ack=64
S8 ,S(m8 =43 )
Ack9 ,S
Seq=64,Ack=210
S7 ,C2(m7 =8 )
Ack8 ,C2
Seq=210,Ack=72
S9 ,S(m9 =617 )
Ack10 ,S
Seq=72,Ack=827
ACK
Seq=827,Ack=72
S10 ,S(m10 =3 )
Ack11 ,S
Seq=72,Ack=830
S8 ,C2(m8 =6 )
Ack9 ,C2
Seq=830,Ack=78
S11 ,S(m11 =35 )
Ack12 ,S
Seq=865,Ack=78
FIN ,ACK
Seq=78,Ack=866
ACK
Seq=78,Ack=866
FIN ,ACK
Seq=866,Ack=79
ACK